

Increasing the Acceptance of Industrial Robots by Adapting Movement Behavior to Individual User Differences

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ABSTRACT

Machines, computers, and robots play an increasingly important role in our personal and professional lives. In manufacturing industries, the direct physical collaboration between workers and robots is already reality today. To understand how humans perceive this collaboration, investigations regarding the effect of different robot behaviors and the characteristics that influence the perception by humans are required. Existing findings on the effect of human attributes and human-like appearance of robots point out several possibilities that improve interactions between humans and robots. However, these studies predominantly focus on humanoid robots. With regard to the increasing number of industrial robots and the arising economic potential of automation, the objective of my research is to gain in-depth understanding of human perception of robot movement behaviors. Achieving this requires insights on possible and perceptible robot behaviors in the non-humanoid field, how they are perceived and how they can be mapped to acceptance by humans. Based on this understanding, resulting interactions can be improved by adapting relevant behavior to the individual's preferences. In view of new technologies and possibilities, it might even be possible to use real time data to classify humans and let robots adapt their behavior automatically.

CCS CONCEPTS

• **Human-centered computing** → **User studies; Empirical studies in HCI**; • **Computer systems organization** → **Robotics**.

KEYWORDS

HRI, Industrial Robot, Movement Behavior, Acceptance, Human-Likeness

ACM Reference Format:

Damian Hostettler. 2022. Increasing the Acceptance of Industrial Robots by Adapting Movement Behavior to Individual User Differences. In *CHI Conference on Human Factors in Computing Systems Extended Abstracts (CHI '22 Extended Abstracts)*, April 29-May 5, 2022, New Orleans, LA, USA. ACM, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3491101.3503812>

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CHI '22 Extended Abstracts, April 29-May 5, 2022, New Orleans, LA, USA

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ACM ISBN 978-1-4503-9156-6/22/04.

<https://doi.org/10.1145/3491101.3503812>

1 INTRODUCTION

Interactions between humans and robots have been investigated for decades. As early as 1970, Mori described the effect of robots that look like or act like humans and investigated specifically how simulated behavior and appearance are non-linearly related to the user's acceptance of a robot [37]. With increasing importance of robots in both private and business contexts, a deep understanding of what makes human-robot interactions satisfactory and how the ergonomics can be further improved is of critical importance, enabling modulation of physical as well as psychological aspects. Since the number of industrial and collaborative robots (cobots) is rising rapidly [26] and since cobots are built for close collaborations with humans and require fewer safety precautions, the optimized design of interactions becomes even more important. However, existing research in the HRI field focuses mostly on appearance and behavior of humanoid and social robots with human characteristics such as gaze, mimics, or voice [39][36][32][27][30][10][2][24][45][5]. Even though some research considers robot gestures, the focus is predominantly on co-verbal gestures, investigated using humanoid robots as well [47]. Human-robot interactions are also investigated from a sociological perspective. In this field, the focus is predominantly the impact of automation technology on industrial work and effects like replacement of jobs by machines [20][7][15]. Regarding industrial robots, specific constructs such as transparency, legibility and predictability when planning ergonomic robot movements have been investigated to let robots make their "intentions clear to its human collaborator" [11][19][8]. Similarly, an emerging line of research has examined hand-over tasks and their effects on physiological parameters [16][9]. Robotic arms are used for several applications in industrial shopfloors such as material handling, packaging, assembly and even complex processes like welding or painting. Other than humanoid robots, these robots have limited human-like characteristics and cannot interact through facial expressions or verbal communication. Since we are unable to modulate such characteristics for industrial cobots, it is critical to advance our understanding on how other factors that we can modulate (e.g., robot arm movements) influence their acceptance. With regards to future potential of understanding such interdependencies, this might enable to let limited devices (without human-type personality affordances, e.g., vacuum cleaners, buildings, computers, etc.) behave differently based on individual preferences. Modern technologies such as emotion recognition, eye tracking or personality traits detection based on individual mobile app adoption [55] might even enable that such limited devices not only choose autonomously between certain behaviors but also detect relevant human characteristics in real time and adapt their behavior accordingly. In summary, during my PhD I propose to investigate how the specific movement behaviors of limited devices – starting with an industrial robot – can be adapted

to individual differences, increasing the acceptance of such devices. To understand how robot movement behavior affects human perception, the following explanations lay out related work in the form of a theoretical foundation and propose a series of experiments as methodical approach.

2 RELATED WORK

Many studies in HRI investigate how interactions between humans and robots can be optimized in terms of acceptance and ergonomics. Findings from interactions among humans serve as framework to understand and design the interplay between people and technical devices. Furthermore, psychology and marketing research has been trying to understand what distinguishes groups of people and to form segments of people for decades, enabling companies and technologies to address specific target groups accurately.

2.1 Human-Likeness of Machines and Robots

Findings on interactions of people with machines and new technologies put forth several theories stating that important effects and characteristics of human-human interactions are true for human-robot/human-machine interactions as well. Establishing media equation theory – i.e., the notion that people humanize technical devices - Reeves & Nass [41] were able to show that people are not only polite to other people but also to computers and that social rules apply to media, even though the computer “showed plain text on a plain black-and-white screen” and did not have any artificial intelligence, virtual reality, or “any other obvious display of social presence”. Furthermore, anthropomorphism describes how people attribute humanness to non-human things and existing findings show that anthropomorphizing is influenced by individual and cultural differences [3] and [31] state that “anthropomorphism is a promising approach to improve the acceptance of a robotic system as a team-partner”. For humanoid robots, considerable research has investigated human-likeness and “people’s responses to human-like robots” [39]. The recently published ABOT Database for example provides a collection of variations for human-like appearances of contemporary robots and scores robots with regard their human-like appearance [56]. However, these studies focus on human-like appearance features such as eyes, arms, etc. Regarding the perception of robot movement behaviors, similar social cognition theories are applied. (De)Humanization [23] as well as mind perception [22][52][51][39] describe how humans perceive others and under which circumstances they accept them and even attribute humanlike mind to even inanimate objects. Following Ruijters et al.’s statement that anthropomorphism “involves the tendency to attribute human characteristics to robots” [46], the above-mentioned theories serve as theoretical foundation for the design of effective robot movement behaviors along actual human behaviors. Based on these theories and the attribution of human-likeness to robots, the study of robot movements can be designed in a way that perceived human-likeness increases, enabling a higher probability of enhanced cooperation, engagement, enjoyable interactions as well as empathy attributions[28][29][42]. In addition, according to similarity attraction theory, people find others more attractive the more similar they perceive them. Therefore, my interest is to also further investigate whether enhanced similarity of an industrial robot and

the human’s characteristics – represented by different robot movement behaviors – leads to attraction or repulsion tendencies[14] and cf. [49] in human-robot interactions.

2.2 Individual Differences that Explain Differing Behaviors and Perceptions

To understand different perceptions of interactions with robots and for the design of effective robot movements, differentiating human characteristics are needed. Basic demographic characteristics such as age, gender or location can be captured easily and might explain some differences in the perception of robots. Additionally, existing research on individual differences and decision-making often investigated personality traits, drawing the attention of the HRI and technology acceptance field to these concepts [17][33][38][6][1][4][25][35][34][44]. Therefore, the Big Five Framework (also FFM/OCEAN model) will be used as an extension to demographic characteristics in order to differentiate homogeneous segments of people with similar perceptions. While Goldberg used the term “Big Five” for the first time in 1981[21], Digman states (even in 1990) that “more than 20 years ago ... the domain could be adequately described by five superordinate constructs”[18]:

- Openness to experience
- Conscientiousness
- Extraversion
- Agreeableness
- Neuroticism

The five personality traits are usually determined using a questionnaire. Numerous studies have tested the relevant items, from which questionnaires with 10 to 240 items were developed [13][40]. More recent research developed systems that assess personality automatically, e.g., based on the installed apps and smartphone usage of the respondent [12][55]. Personality research in the HRI field has put forth several findings on the interplay between humanoid robots and personality traits. In addition to human characteristics (gaze, mimics etc.), these studies indicate that movement behavior of humanoid robots affect the acceptance of robots as well. Correspondingly, these studies demonstrated that people high in extraversion preferred high frequency of movements, expansive gestures, high motion speeds and motion angles whereas introverted people preferred backward leaning, turning away, narrow gesture amplitudes and inward gesture directions [54][43][32][38]. As described above, my interest is to transfer these insights from humanoid robots to industrial robots and movement behaviors only. For the experimental approach described below, personality therefore serves as a supplement to demographic variables and additional criterion when investigating individual differences and their effects on the perception of human-robot interactions as well as support for the design of movement characteristics.

3 METHODOLOGY: EXPERIMENTAL INVESTIGATION OF HUMAN-ROBOT INTERACTIONS

To investigate the relationship between robot movements and the perception of these movements, depending on individual differences, we plan to conduct a series of online experiments as a first

step. Regarding video-manipulations, it is important to note that social cognition processes “are so pervasive and spontaneous that they take place not only in face-to-face interactions, but also in technology mediated settings” [14] and that video-based HRI methods are equally valid while being less expensive and with the possibility to utilize larger samples, with greater statistical power, in a shorter time[50]. To confirm initial findings from these online-experiments, lab experiments as well as studies with actual robot/machine operators will follow. Using manipulated robot behaviors as an independent variable (IV), the perception of these behaviors as dependent variable (DV) and individual differences as moderators (MV), these variables will be conceptualized as follows.

3.1 IV: Manipulation of Robot Behavior

Videos of a robotic arm serve as stimulus showing different movement parameters such as speed, part approach, rotations, smoothness, moving range and approach direction in different conditions each. For the manipulation checks and experiments carried out so far, a UR 10e robot was used. Regarding effective movement behaviors, existing findings from humanoid robots are used and adapted to an industrial robot’s movement capacities using both, personality as well as demographic characteristics. As mentioned before, people with different personality traits prefer different movement behaviors such as broad/narrow gestures and females as well as people high in neuroticism prefer to stand farther from robots [48].

3.2 DV: Measuring Perception and Acceptance based on Human-Likeness

A number of validated scales are used to measure the dependent variable. Scales such as the ALMERE model [24] and the Godspeed Questionnaire Series [53] provide several items that are used to measure the user’s experience and further cover the perception of the above-mentioned human characteristics.

3.3 MV: Capturing Individual Differences

Demographic characteristics include age, gender, prior experience with robots, location and education and will be asked directly. Personality will be measured using the standardized personality test IPIP-50 developed by Goldberg in 1992[21].

4 STATE OF OWN RESEARCH

During the last months, checks with two different sets of stimuli have been conducted. Based on existing literature and team discussions, different robot movements have been classified and analysed to determine high and low expressions of the movement parameters, having regard to the UR-10e’s degrees of freedom. The first structure included 8 different movement characteristics, grouped into 3 main parameters (see tables 1-3) with three conditions each. Videos were recorded showing the robot either in low or high condition for one of the parameters while the other two parameters were left in neutral condition, resulting in six different movement behaviors. These combinations were tested for significant mean differences with random assignment of all participants to one of the conditions per parameter. Even though some parameters reached significance the results of two independent manipulation checks with 120 participants in each case (20 per condition) showed several emerging

inconsistencies, probably due to overlaps of the movement parameters. Participants were recruited using MTurk for Check1 and Prolific for Check2. The second, revised stimuli structure was reduced to 6 separate movement parameters without grouping and a maximum variation resulting in two conditions per parameter (see table 4).

This revised structure was tested again with 80 participants (40 per condition) and random assignment per video of all participants to one of the conditions, answering if they perceive the robot to be slow or fast, with care or without care, calm or nervous etc. In addition, participants answered open questions about their perception of the robot movements. Surprisingly, only the perception of the two conditions for “approach direction” reached statistically significant differences between means with $t(79) = -9.56, p < 0.001 (M1 = 2.48, M2 = 4.59)$. No other effects reached significance with all $p's > 0.1$. Since the two conditions for all manipulations except “Approach Direction” were not perceived as significantly different, the following preliminary summary of findings can be derived:

- Novices cannot assess the robot movements due to missing baselines regarding the parameters mentioned above.
- Participants (as well as the author) might linguistically misinterpret the naming and gradation (e.g., very careful vs. very careless) of the parameters and/or conditions.
- The results do not take moderators into account, these might possibly explain different perceptions based on e.g., demographic criteria or personality traits.

Therefore, a third manipulation check consisting of a combination of manipulation check and moderators will be conducted. In addition, perceived human-likeness will be tested using the Godspeed anthropomorphism questionnaire. The underlying assumption is that taking into account individual differences might deliver a better explanation for the weak effects from Check 1 and Check2.

5 NEXT STEPS

Since only “approach direction” using stimuli structure 2 reached significant differences between conditions 1 and 2, an additional manipulation check including the above-mentioned moderators will be carried out. The results are expected to show more differences between the perceptions in condition 1 and 2 when separating the participant’s answers by demographic variables (and personality traits in a second step).

1. Randomised assignment of the conditions to the participants resulting in 6 videos per participant and questions regarding the perception of the different conditions
2. Questions on demographic characteristics of the participants
3. 50 participants per condition resulting in $n=100$, recruited using Prolific
4. Each participant takes the standardized IPIP 50 personality test

Once the manipulations have been shown to be effective, an experiment the participant’s preference for each condition will be investigated, depending on the participant’s personality and demographic characteristics.

1. Each participant takes the standardized IPIP 50 personality test

Table 1: Performance Parameters

	Low	Neutral	High
Robot Speed	60%	80%	100%
Stressed vs. Smooth movements	Linear movements; quick stops between coordinates	Steady movements, no smoothing of curves	Steady movements; smoothing of curves
Unconventional vs. Ordinary movements	Unnecessary rotations, movements behind and below robot's center	Economical movements without attention to robot's center	Economical movements in front and above robot's center

Table 2: Precision Parameters

	Low	Neutral	High
Slow vs. fast approach	Speed remains equal to movements; gripper opens while robot in motion	Speed reduction by 25%; gripper opens just above artifact	Speed reduction by 50%; brief stop before gripper opens
Small vs. large distance between artifacts	Distance = Artifact + Gripper width + 30mm	Distance = Artifact + Gripper width + 10mm	Distance = Artifact + Gripper width
Gripper opening width	Artifact + 10mm	Artifact + 30mm	Artifact + 50mm

Table 3: Spatial Parameters

	Low	Neutral	High
Inward vs. broad movements	Maximum shoulder/elbow joints extension = 40%	Maximum shoulder/elbow joints extension = 70%	Maximum shoulder/elbow joints extension = 100%
Approach direction	Movements to the side, approaches participant from right	Movements to the side, approach directly towards participant	Movement straight ahead towards the participant

Table 4: Revised Stimuli Structure

Parameter	Condition 1	Condition 2
Speed	Slow	Fast
Part Approach	With care	Without care
Smoothness	Calm	Nervous
Rotation	Conventional rotations	Unconventional rotations
Movement Range	Small movement range	Large movement range
Approach Direction	From the side	Straight towards you

2. Randomised assignment of the conditions to the participants resulting in 6 videos per participant and a quantitative question regarding the preference/satisfaction with the condition
3. Questions on demographic characteristics of the participants
4. 150 participants per characteristic (personality trait) resulting in n=750, recruited using Prolific

Using the findings on participant's preferences, matching robot movement behaviors will be designed corresponding to relevant personality traits and demographics. Participants will then be asked to rate their perception of the movement behaviors using different items to measure acceptance and perceived human-likeness as dependent variable.

1. Each participant takes the standardized IPIP 50 personality test
2. Randomised assignment of the participants to the videos regarding the preference/satisfaction with the condition
3. Several items assessing the perception and evaluation of the robot shown in the video
4. Questions on demographic characteristics of the participants
5. 150 participants per characteristic (personality trait) resulting in n=750, recruited using Prolific

On this basis, additional lab experiments will follow to confirm findings in face-to-face interactions with real robots. Insights on the interplay of individual differences and corresponding preferences for certain movement behaviors will in the medium term be used

to build a prototypical cyber physical system that detects individual differences and adapts its behavior automatically. Performing additional tests with actual machine operators and shopfloor workers will show the effect of such a system in the real life of people working with industrial robots. Following the chain of arguments presented above, this should result in increased acceptance and allow to transfer these findings to other devices in future.

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